IBM SkillsBuild

Data Analytics with AI Academic Internship Program

BharatCares in association with AICTE

PROJECT REPORT

YouTube Trending Videos Analysis: What Makes a Video Go Viral?

Submitted by

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Tools: Python, pandas, matplotlib, seaborn, Google Colab, IBM Bob

Dataset: YouTube Trending Videos Dataset (The Devastator, Kaggle)

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1. Introduction

YouTube is one of the largest video platforms in the world, and thousands of creators compete every day to reach a wider audience. Every day YouTube also publishes a "Trending" list of videos that are getting unusually high attention. Understanding what these trending videos have in common can help new creators plan better content.

This project analyses a public dataset of YouTube trending videos using Python. The analysis looks at which channels and categories attract the most views, whether the day and time of publishing matter, how likes, comments and views relate to each other, and whether disabling comments affects a video's reach. The project was completed as part of the IBM SkillsBuild Data Analytics with AI Academic Internship Program, with IBM Bob used as an AI coding assistant to help write and structure the notebook.

2. Problem Statement and Objectives

Problem statement: What patterns in channels, categories, publishing time and audience engagement are associated with higher views on YouTube trending videos?

Objectives of the project:

- Load, explore and clean a real-world YouTube trending videos dataset.
- Create new engagement measures (like ratio and engagement rate) from the existing columns.
- Identify the top channels and the best-performing video categories by views.
- Find out whether the publishing day and publishing hour are linked with higher average views.
- Measure the relationship between views, likes, dislikes and comments.
- Compare videos with comments enabled and comments disabled.
- Present clear findings and practical recommendations for content creators.

3. Dataset Description

The dataset used is "YouTube Trending Videos Dataset" by The Devastator, downloaded from Kaggle as a single CSV file (about 79 MB). It contains 18 columns, and after cleaning the analysis used about 122,760 video records. The most important columns are described below.

Column	Description
video_id, title	Unique video identifier and the video title
channel_title	Name of the channel that uploaded the video
category_id	Numeric YouTube category of the video (for example 10 = Music, 24 = Entertainment)
trending_date, publish_date	Date the video trended and the date it was published
time_frame	One-hour slot in which the video was published (for example 4:00 to 4:59)
published_day_of_week	Day of the week on which the video was published
views, likes, dislikes, comment_count	Numeric engagement counts for each video
comments_disabled	Whether the uploader turned off comments on the video
tags	Keywords added by the uploader (blank values were filled with "none")

Summary statistics show that the data is highly skewed. The average video has about 2.82 million views, but the median video has only about 391,000 views, and the most viewed video has about 424.5 million views. A small number of extremely popular videos pull the averages upwards, which is important to remember when reading the results.

4. Tools and Technologies

Tool	Purpose
Python 3	Main programming language for the analysis
pandas, numpy	Loading, cleaning, grouping and calculating statistics
matplotlib, seaborn	Creating charts and the correlation heatmap
Google Colab	Running the Jupyter Notebook in the cloud
IBM Bob	AI assistant used to generate and refine the notebook code, requirements file and README
Kaggle	Source of the dataset

5. Methodology

5.1 Data Cleaning

- Checked for duplicate rows: 0 duplicates were found before and after cleaning.
- Checked for missing values: blank tags were filled with the text "none", and rows with missing views, likes, dislikes or comment count were dropped.
- Converted trending_date (format YY.DD.MM) and publish_date into proper date types.
- Removed columns that were not useful for the analysis.

5.2 Feature Engineering

Two new columns were created to measure how strongly viewers react to a video: like_ratio, and engagement_rate, calculated as (likes + comments) divided by views. A readable comments_disabled_label column (Comments Enabled or Comments Disabled) was also created for the comments analysis.

5.3 Analysis Approach

Six analyses were carried out: top channels by total views, average views by category, by publishing day and by publishing hour, the relationship between likes and views (scatter plot and correlation heatmap), and the effect of disabling comments. Every chart in this report was produced in the project notebook (Sakshi_YouTubeTrendingAnalysis.ipynb).

6. Analysis and Results

6.1 Top 10 Channels by Total Views

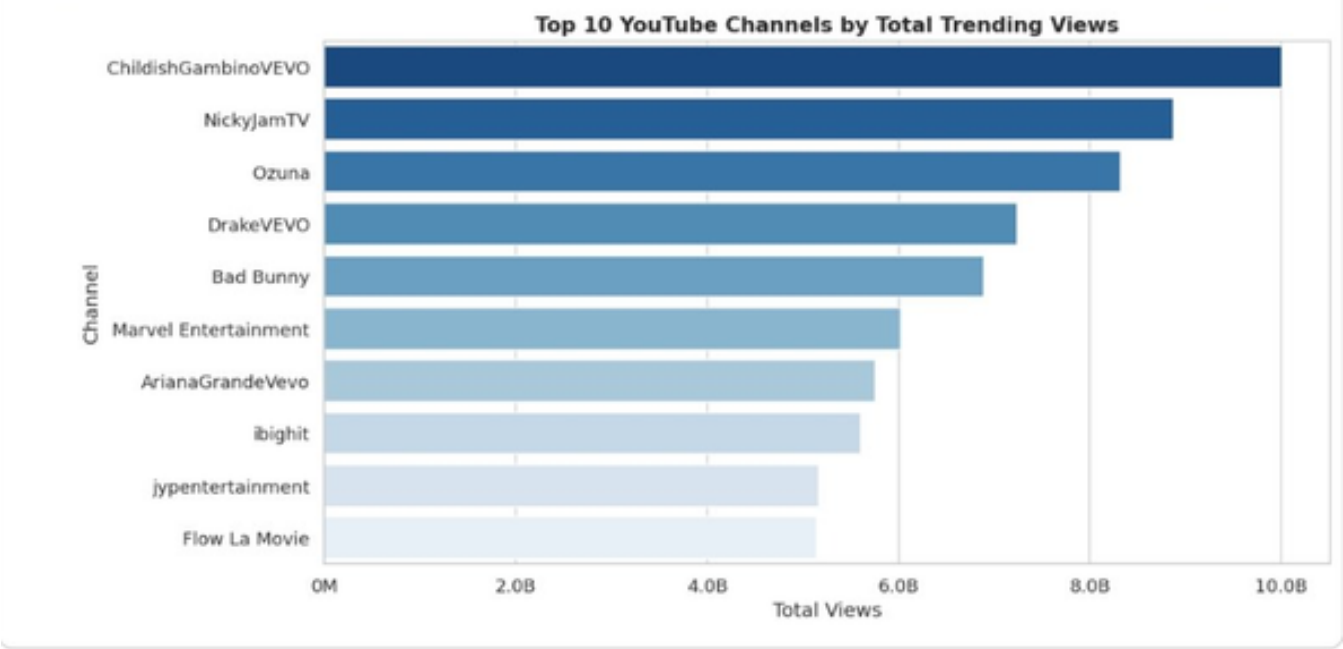


Figure 1: Top 10 YouTube channels by total trending views

ChildishGambinoVEVO is at the top with roughly 10 billion total trending views, followed by NickyJamTV, Ozuna, DrakeVEVO and Bad Bunny. Almost all channels in the top 10 belong to music artists or music labels (for example Ariana Grande, ibighit and jypentertainment); Marvel Entertainment is the main non-music channel.

Insight: A small group of very large channels collects a huge share of the total views, and music dominates the list.

6.2 Average Views by Category

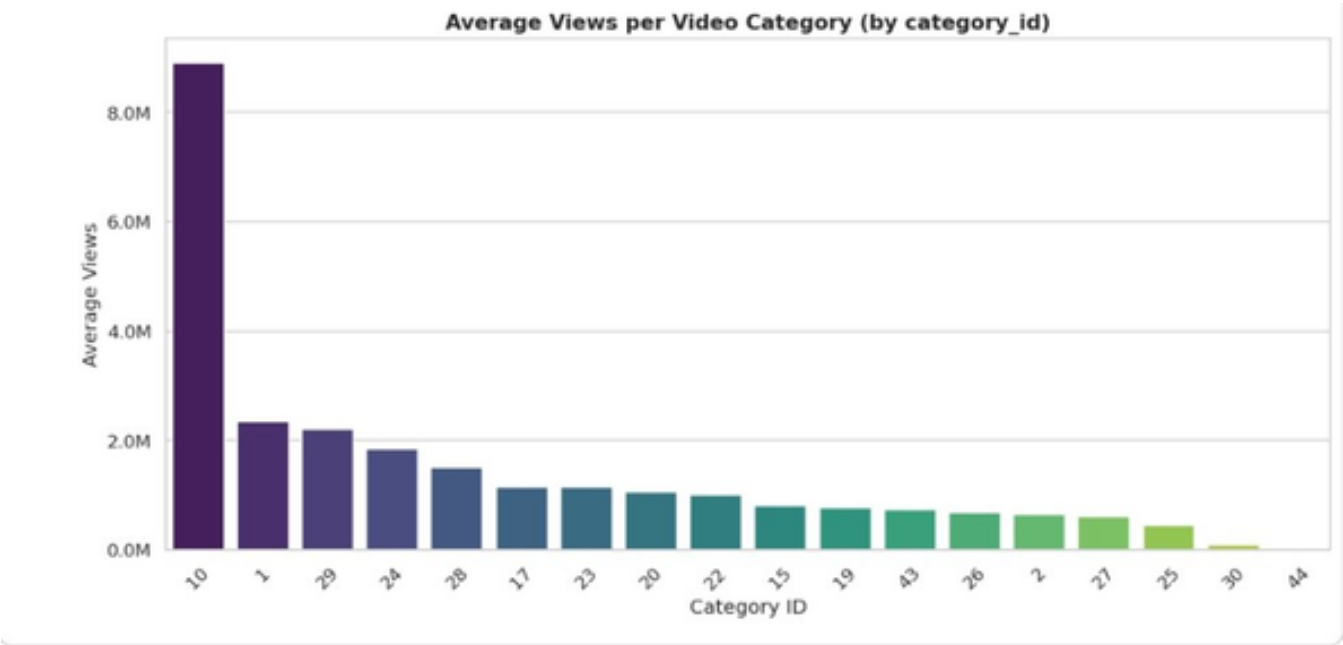


Figure 2: Average views per video category (by category_id)

Category (ID)	Standard YouTube name	Average views
10	Music	8,905,893
1	Film & Animation	2,340,189
29	Nonprofits & Activism	2,205,240
24	Entertainment	1,850,044
28	Science & Technology	1,502,168

Insight: Category 10 (Music) is far ahead of every other category, with an average that is almost four times higher than the second-placed category. Category choice has a large influence on potential reach.

6.3 Best Day of the Week to Publish

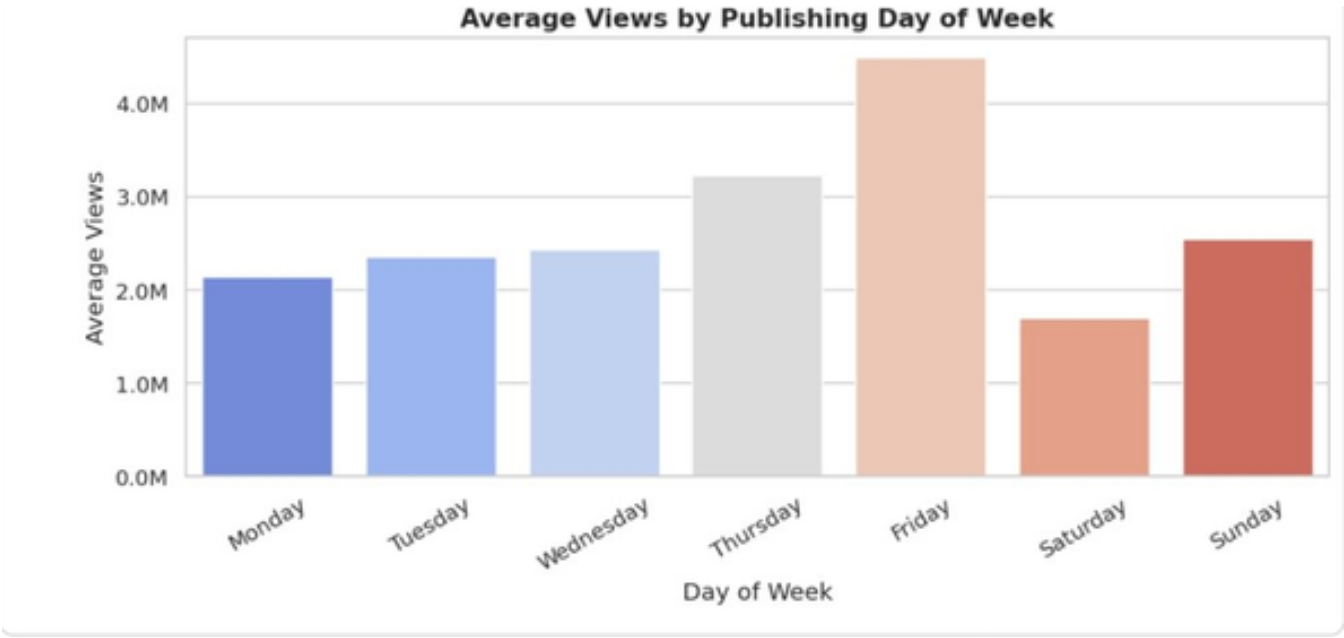


Figure 3: Average views by publishing day of the week

Day	Average views
Monday	2,138,786
Tuesday	2,356,786
Wednesday	2,432,047
Thursday	3,231,785
Friday	4,485,613
Saturday	1,696,119
Sunday	2,547,532

Insight: Friday has the highest average views (about 4.49 million) and Saturday has the lowest (about 1.70 million). Views rise steadily from Monday to Friday, then drop sharply on Saturday.

6.4 Best Publishing Time

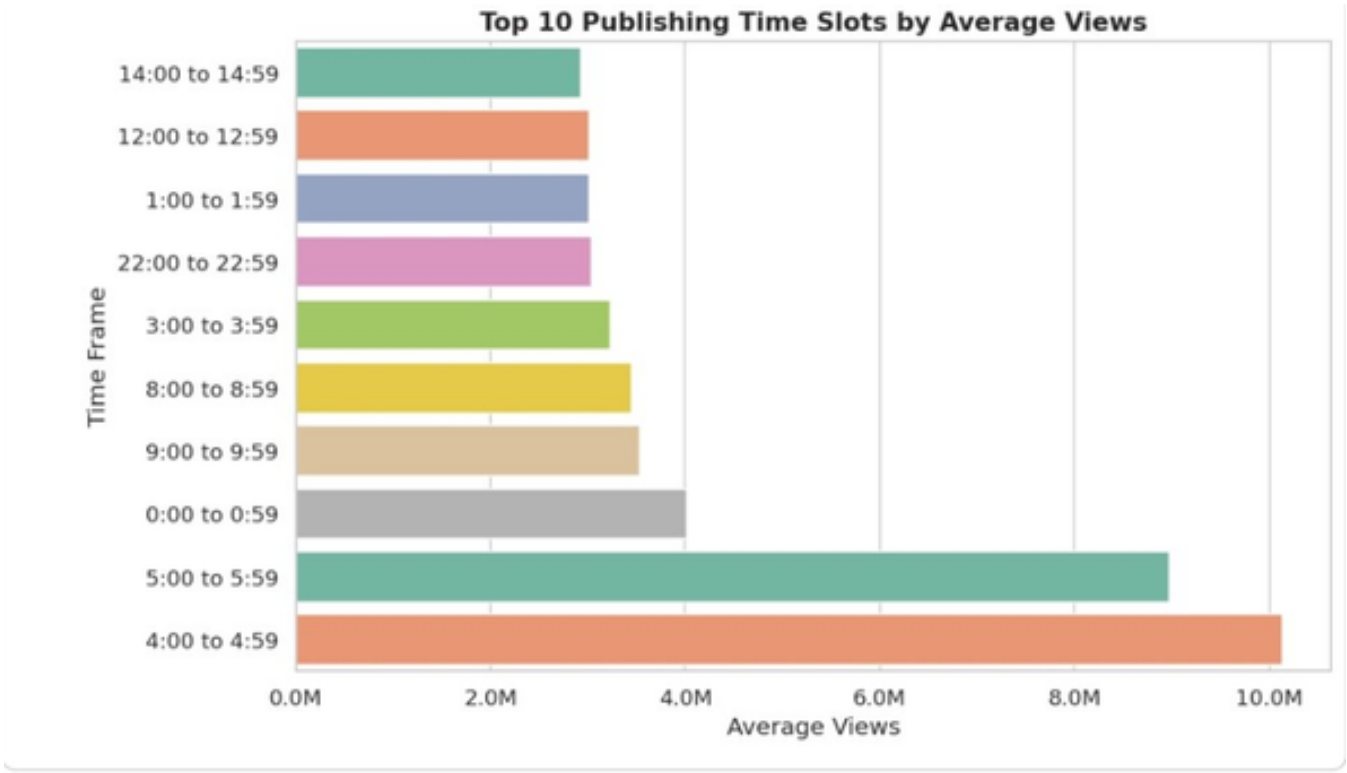


Figure 4: Top 10 publishing time slots by average views

Time slot	Average views
4:00 to 4:59	10,131,389
5:00 to 5:59	8,972,228
0:00 to 0:59	4,024,025
9:00 to 9:59	3,540,727
8:00 to 8:59	3,452,512

Insight: The 4 AM and 5 AM slots have the highest average views. Because averages are strongly affected by a few extremely popular videos, these peaks may be driven by a handful of viral uploads, so they should be treated with caution rather than as a guaranteed best time to publish. The time zone of the publishing times is not stated in the dataset.

6.5 Likes vs Views and Correlation Analysis

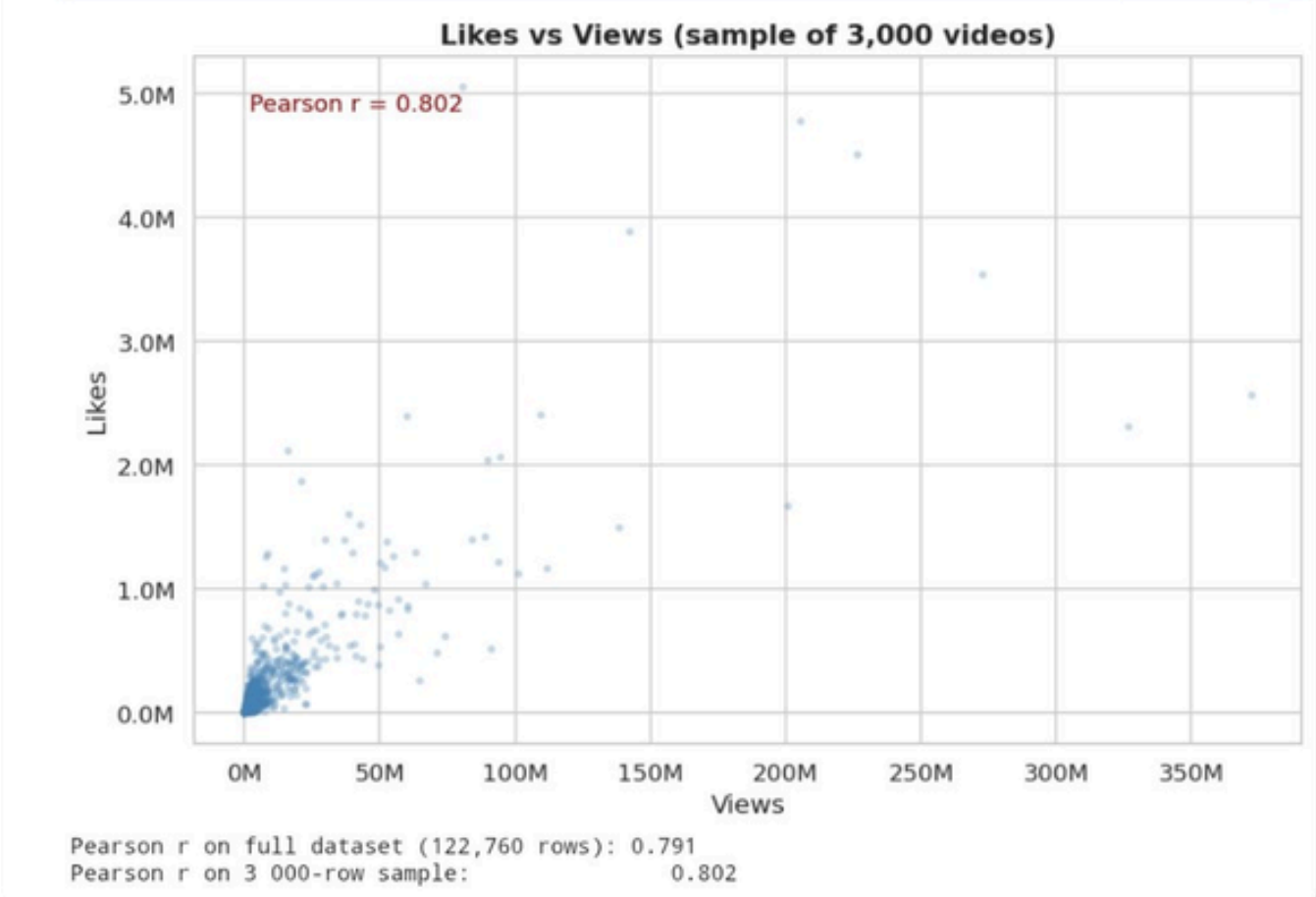


Figure 5: Likes vs views scatter plot (sample of 3,000 videos)

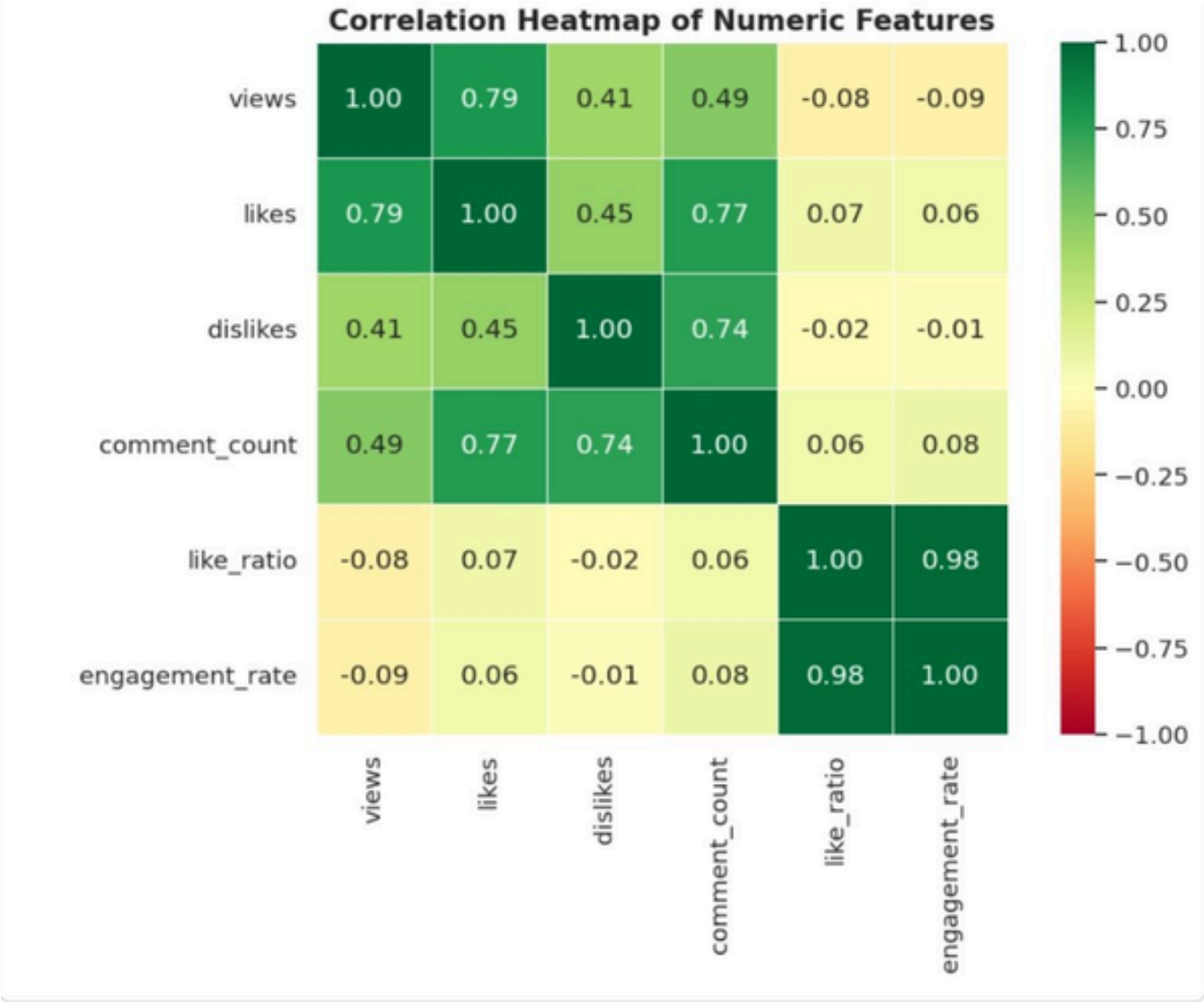


Figure 6: Correlation heatmap of numeric features

The Pearson correlation between views and likes is 0.79 on the full dataset of about 122,760 rows (0.80 on the 3,000-row sample used for the scatter plot). This is a strong positive relationship. Other relationships from the heatmap are:

- Views and comment count: 0.49 (moderate positive).
- Views and dislikes: 0.41 (moderate positive, since popular videos attract all kinds of reactions).
- Likes and comment count: 0.77, and dislikes and comment count: 0.74 (strong).
- Views and the two ratio measures (like_ratio and engagement_rate): about -0.08 and -0.09, meaning a video with more views does not necessarily have a higher engagement rate.

Insight: Videos that get more views also tend to get more likes and comments, but the relationship is strong rather than perfect (r is below 0.9), and correlation does not prove that one causes the other.

6.6 Effect of Disabling Comments

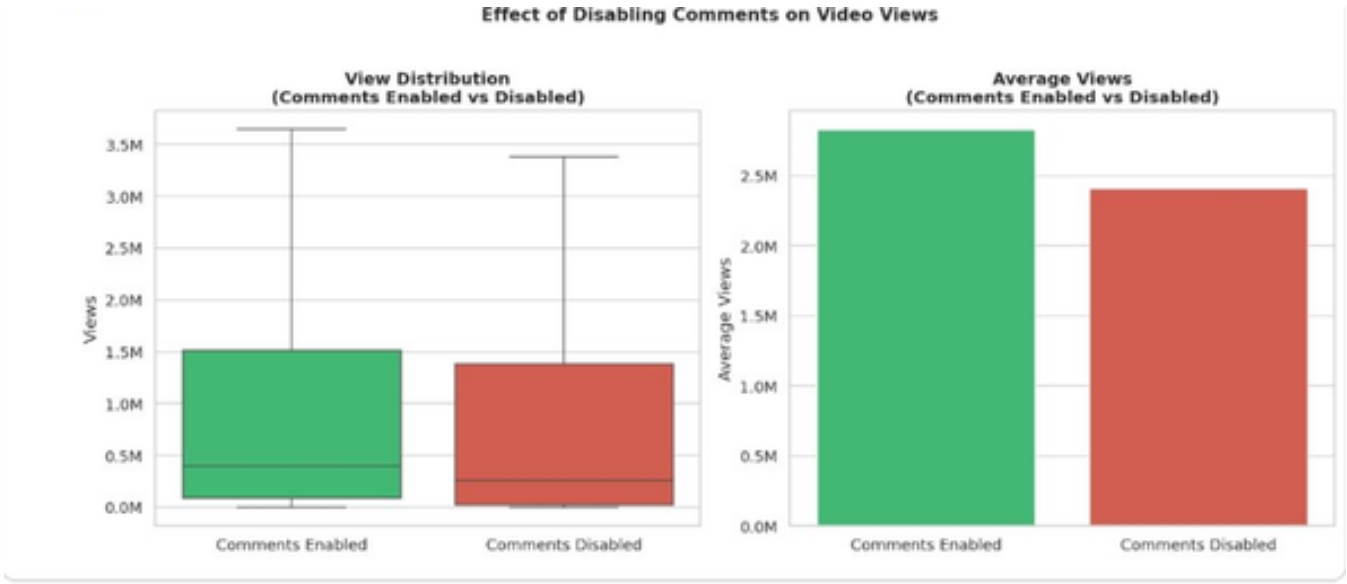


Figure 7: Views of videos with comments enabled vs disabled

Group	Videos	Share
Comments Enabled	120,527	98.2%
Comments Disabled	2,233	1.8%

Insight: Videos with comments enabled show higher median and average views than videos with comments disabled. However, only 1.8% of the videos have comments disabled, so the averages of that group are less stable and the comparison is not perfectly balanced.

7. Key Findings

#	Finding
1	A few mega-channels dominate total views; ChildishGambinoVEVO leads with roughly 10 billion trending views, and most top-10 channels are music channels.
2	Category matters enormously: Music (ID 10) averages about 8.9 million views per video, far above Film & Animation (2.34M), Nonprofits & Activism (2.21M), Entertainment (1.85M) and Science & Technology (1.50M).
3	Friday is the best publishing day (about 4.49M average views) and Saturday is the weakest (about 1.70M).
4	Early-morning slots (4:00 to 5:59) show the highest average views, but this may be influenced by a few viral videos.
5	Likes and views are strongly correlated (r = 0.79 on the full dataset); comments are moderately correlated with views (r = 0.49).
6	Videos with comments enabled have higher median and average views, but the comments-disabled group is very small (1.8%).

8. Recommendations for Content Creators

- **Choose the category carefully.** Music, Film & Animation and Entertainment content reached the largest average audiences in this dataset, so creators should consider how their topic fits these categories.
- **Plan the publishing day.** Friday uploads had the highest average views, while Saturday had the lowest, so schedule important videos before the weekend.
- **Experiment with timing.** Test different upload times for your own audience instead of relying only on the early-morning peak, which may be caused by a few outlier videos.
- **Keep comments enabled and encourage engagement.** Open comment sections were linked with better view counts, and likes and views move together, so encourage viewers to like and comment early.
- **Learn from top channels.** The biggest channels post consistently with strong titles and thumbnails; studying them can give ideas for your own niche.

9. Limitations

- The dataset contains only videos that reached the trending list, so it cannot show what makes videos fail to trend.
- Averages are heavily affected by a few extremely popular videos (the mean views are far above the median views).
- The comments-disabled group is small (1.8%), so conclusions about it are less reliable.
- The data is a historical snapshot, so patterns may differ on today's YouTube.
- Correlation does not prove causation: the analysis shows associations, not guaranteed results.

10. Conclusion

This project used Python and a real YouTube trending dataset of about 122,760 videos to explore what is associated with high views. The results show that category, channel size, publishing day and audience engagement are all linked with video performance, with Music content, Friday uploads and strong like activity standing out. At the same time, the limitations show why these results should be used as guidance and tested by creators on their own channels. The project also gave practical experience in data cleaning, feature engineering, visualisation and insight writing, and showed how an AI assistant such as IBM Bob can speed up development when its output is carefully checked against the real data.

11. Project Files and References

GitHub repository:

GitHub - raisakshiii941-ux/you-tube-trending--Analysis: YouTube Trending Videos Analysis using Python, pandas ...

Dataset: YouTube Trending Videos Dataset by The Devastator, Kaggle (kaggle.com/datasets).

Submitted files: Sakshi_YouTubeTrendingAnalysis.ipynb, requirements.txt, Sakshi_ProjectReport.docx and README.md.

References: pandas, matplotlib and seaborn official documentation; IBM SkillsBuild Data Analytics with AI course material.